INCH-POUND
MIL-DTL-24643/80
27 November 2023

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 1,000 VOLTS, SINGLE-CONDUCTOR, TYPE LST AND TYPE LSTW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LST (non-watertight) and LSTW (watertight).

Construction (Non-Watertight and Watertight)

First - Copper conductor, coated or uncoated, in accordance with ASTM B8 or Navy Standard (see table I or table II for size and number of strands).

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Thermoset insulation (see table I or table II for thickness).

NOTE 1: Finished wire used as components in other specification sheets shall not be printed to this specification sheet. Finished wire not used as components in other specification sheets shall be printed in accordance with 3.4.10 of MIL-DTL-24643, without the jacket type information. The printed characters may be parallel or perpendicular to the longitudinal axis of the wire. Wire sized smaller than 0.125 inch shall be sized at 0.047 inch character height.

NOTE 2: Manufacturer's identification tape (3.4.9 of MIL-DTL-24643) is not applicable.

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details (non-watertight).

Military part no. M24643/80	Type and size	Conductor size (AWG/MCM)	Number of strands	Insulation thickness min. avg. (inch)	Wire diameter (inch)		Conductor resistance (ohms)	Watertightness max. (cubic inches)
					min.	max.		
-01UN	LST-2	18	7	0.027	0.105	0.110	6.87	N/A
-02UN	LST-3	16	7	0.027	0.115	0.120	4.27	N/A
-03UN	LST-4	14	7	0.027	0.125	0.135	2.68	N/A
-04UN	LST-6	12	7	0.027	0.145	0.155	1.69	N/A
-05UN	LST-9	10	7	0.030	0.175	0.190	1.06	N/A
-06UN	LST-14	9	7	0.042	0.215	0.230	0.843	N/A
-07UN	LST-23	7	7	0.042	0.245	0.265	0.534	N/A
-08UN	LST-40	4	19	0.042	0.315	0.335	0.263	N/A
-09UN	LST-50	3	19	0.042	0.345	0.365	0.209	N/A
-10UN	LST-60	2	37	0.045	0.380	0.405	0.165	N/A
-11UN	LST-75	1	37	0.052	0.430	0.460	0.132	N/A
-12UN	LST-100	0	61	0.052	0.465	0.495	0.105	N/A
-13UN	LST-150	000	61	0.052	0.510	0.550	0.0663	N/A
-14UN	LST-300	300	91	0.060	0.735	0.775	0.0370	N/A
-15UN	LST-400	400 (Navy STD)	127	0.060	0.870	0.910	0.0268	N/A

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TABLE II. Details (watertight).

Military part no. M24643/80	Type and size	Conductor size (AWG/MCM)	Number of strands	Insulation thickness min. avg. (inch)	Wire diameter (inch)		Conductor resistance (ohms)	Watertightness max. (cubic inches)
					min.	max.		
-16UN	LSTW-2	18	7	0.027	0.105	0.110	6.87	1
-17UN	LSTW-3	16	7	0.027	0.115	0.120	4.27	2
-18UN	LSTW-4	14	7	0.027	0.125	0.135	2.68	2
-19UN	LSTW-6	12	7	0.027	0.145	0.155	1.69	3
-20UN	LSTW-9	10	7	0.030	0.175	0.190	1.06	4
-21UN	LSTW-14	9	7	0.042	0.215	0.230	0.843	5
-22UN	LSTW-23	7	7	0.042	0.245	0.265	0.534	6
-23UN	LSTW-40	4	19	0.042	0.315	0.335	0.263	8
-24UN	LSTW-50	3	19	0.042	0.345	0.365	0.209	9
-25UN	LSTW-60	2	37	0.045	0.380	0.405	0.165	9
-26UN	LSTW-75	1	37	0.052	0.430	0.460	0.132	9
-27UN	LSTW-100	0	61	0.052	0.465	0.495	0.105	9
-28UN	LSTW-150	000	61	0.052	0.510	0.550	0.0663	10
-29UN	LSTW-300	300	91	0.060	0.735	0.775	0.0370	11
-30UN	LSTW-400	400 (Navy STD)	127	0.060	0.870	0.910	0.0268	11

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I or table II
Voltage withstand (volts, root mean square, min.)	
Conductor to ground	
18 AWG through 10 AWG	2,500
9 AWG and larger	5,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to ground	
18 AWG through 10 AWG	5,000
9 AWG and larger	2,500

Group A:

Visual and dimensional	No failure
Watertightness (type LSTW only, see table II for limits of water leakage)	No failure

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Drip (203±1.8 °F [95±1 °C]) (type LSTW only) (see MIL-DTL-24643 for limits of drip allowance)	No failure
Physicals (unaged)	
Insulation	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150

Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2) (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75

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Permanence of printing (insulation) (cycles, min.)	125

Group D:

Flame propagation (cable)	Not applicable

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Acid gas equivalent (percent, max.)	
Insulation	18
Halogen content (percent, max.)	
Insulation	0.2
Smoke index (max.)	
Insulation	45
Toxicity index (max.)	
Insulation	1.5
Weathering	Not applicable
Electrical moisture absorption (extruded insulation only)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
All	250

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2017-001)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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